

2.3 Producing robust programs

Why this unit is taught the way it is

What this unit teaches, and where it ends

7 lessons: 6 with an online experience, 1 paper assessment and 0 bonus challenges. 86 activities in all, worth 131 marks.

By the end of it a pupil should be able to:

- know what is meant by defensive design when writing programs **rp-101**
- understand why input validation is necessary **rp-101**
- know a range of validation techniques used to write robust programs **rp-101**
- know what is meant by defensive design considerations **rp-102**
- know a range of problems that can occur when a program is running **rp-102**
- understand some authentication techniques programmers use **rp-102**
- know why creating code that is easy to read is important with large projects **rp-103**
- understand what programmers can do to make their code more readable **rp-103**
- understand how to refine algorithms to make them more robust **rp-103**
- know four reasons why a program should be tested **rp-104**
- know what iterative testing is **rp-104**
- know what final/terminal testing is **rp-104**
- know what a syntax error is **rp-104**
- know what a logic error is **rp-104**
- understand that just because a program works doesn't mean it works for all inputs **rp-105**
- understand that test data should include normal, boundary, invalid and erroneous data **rp-105**
- understand how robust programs are made **rp-106**
- know which lessons I'm secure on and which to revise **rp-106**

Why the lessons are in this order

Each lesson's starter retrieves the one before it, so the order is the retrieval schedule as much as the teaching one. The chain for this unit:

LESSON	TEACHES	RETRIEVES
1 Security desk	know what is meant by defensive design when writing programs	nothing before it in this unit
2 Cyber defence HQ	know what is meant by defensive design considerations	Security desk
3 Code clinic	know why creating code that is easy to read is important with large projects	Cyber defence HQ
4 Bug hunt lab	know four reasons why a program should be tested	Code clinic
5 Test lab	understand that just because a program works doesn't mean it works for all inputs	Bug hunt lab
6 Revision HQ 2.3	understand how robust programs are made	Test lab
7 End-of-topic test	assesses the whole unit	Revision HQ 2.3

The assessment in this unit is placed where it is so that the reflection sheet still has lessons left to send a pupil back to.

What pupils find hard here

None of this unit's walls names a common mistake in so many words. The misconceptions it works against are in the wrong options of its questions instead: every one of those is a plausible wrong idea rather than filler, and the lesson plans list them question by question.

Every multiple-choice and Predict activity in the unit — 77 of them — offers wrong answers that are the misconception, not filler. The split across those options at the hinge question is the cheapest diagnostic in the lesson.

Where the explaining, modelling and practising happen

PHASE WHERE IT IS

Retrieve	The starter on the worksheet, before any device is opened. It is answered on paper and corrected before the lesson moves on.
Explain	The slides and the station walls carry the same explanation, so a teacher can teach from the front and a pupil can re-read it in the room.
Model	There is no worked example on screen in this unit: its activities are questions to answer rather than techniques to copy. The modelling is yours, from the slides, before anyone starts - and it is the part to protect when the lesson runs short.
Practise	78 guided activities, then 8 independent ones.
Check	The badge questions mark as the pupil goes; the hinge question is for the class at once.
Apply	0 challenge activities, and the exam practice on the worksheet.
Review	The exit question and the confidence columns. The confidence columns supplement the evidence; they do not stand in for it.

How the website, the slides and the worksheet divide the work

They are not three copies of the same thing.

- **The slides** are for the front of the room: the explanation and the model, before anyone is working alone.
- **The experience** is where a pupil reads, decides and is marked. It gives the immediate feedback a teacher cannot give thirty pupils at once.
- **The worksheet** is the record and the writing. The key fact in their own words, the station challenge, the exam practice. It is also what survives when the devices do not.

The rule that holds them together is **write first, then answer at the badge**. A pupil who taps first has turned the lesson into a quiz game, and the worksheet is how you can see at a glance who did.

Why a 360° room, and what it should produce

The room is not there to be impressive. It is there because this unit's content is a set of related parts that a pupil has to hold together, and putting one part on each wall makes the set visible and navigable: a pupil can turn back to station 2 while answering station 4. The evidence it should produce is on paper — 18 key facts and 24 station challenges in a pupil's own words — and in the score screen, which shows which station a pupil lost marks on rather than one number.

No claim is made here that immersion improves learning on its own. What the format does is make the structure of a topic physical and keep a pupil's attention on one part at a time; the teaching is in the explanation, the example and the practice, as it would be anywhere else.

Support, and how it is taken away

- The worked example is on the early activities of a station and not on the late ones. That fade is deliberate.
- The hint is a ladder asked for one rung at a time, and the top of it is still the technique on different data, never the answer.
- Read aloud is on every task.

- Prescribed values are boxed and in a monospace face as well as coloured, so they survive a black-and-white printout and a high-contrast mode.
- **Lower literacy and EAL:** pre-teach the vocabulary (Defensive design, Anticipating misuse, Authentication, Input validation, Maintainability, Naming conventions, Iterative testing, Final testing, Syntax error, Logic error, Testing, Logical error). Accept the key fact spoken before it is written, and in two or three words.
- **SEND:** pace is not fixed. Progress saves after every answer and reopens at the earliest unfinished question, so a pupil may finish a station next lesson without losing anything.
- A headset is never required.

Checking, and what to do with what you find

Three checks, in this order: the starter (did last lesson stick), the hinge question (is the class ready for independent work), the exit question (did this lesson stick). The first two change what you do in the lesson; the third changes what you start the next one with.

When a hinge question splits the class, go back to the station it sits on and work its example again, aloud, before anyone goes on. When a test comes back, the reflection sheet names the topics to revisit and review mode on those lessons reopens exactly the questions a pupil got wrong.

Where the approach comes from

The shape of a lesson here — retrieve, explain, model, practise with support that fades, check, apply, review — is ordinary evidence-informed practice rather than anything this product invented. Useful starting points, none of them about Revise 360:

- Rosenshine, *Principles of Instruction* (American Educator, 2012) — the daily review, small steps, worked examples and checking for understanding that this structure follows.
- Sweller, van Merriënboer and Paas on cognitive load and the worked-example effect — why the example comes before the independent task and why the support is withdrawn gradually.
- Dunlosky et al., *Improving Students' Learning With Effective Learning Techniques* (2013) — the evidence for retrieval practice and spacing, which the starters and the review mode use.
- Wiliam, *Embedded Formative Assessment* — hinge questions and what to do with the answers.

What is not claimed. Nothing above is evidence that Revise 360 improves outcomes. No trial of this product has been run, and immersion is not claimed to improve learning by itself. The citations are for the teaching structure; whether these resources work in your classroom is a question your own assessment answers.

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